

*Maximum Number of Names.* Some managers will limit the number of names in the portfolio so that they can better manage the portfolio. These portfolio managers usually employ a combination of quantitative and qualitative portfolio management. These managers perform a quantitative portfolio optimization limiting the number of names to hold in the portfolio, e.g., only hold, say, fifty or fewer names at most. The portfolio manager will follow these limited number of companies in fine detail and make changes if a company has fallen out of favor or if their expectation on potential company growth or dividend stream has changed. It is much more difficult to perform fundamental analysis on a portfolio of several hundred names than on a portfolio with only fifty or fewer names.

*Reflect Future Views.* Managers may specify minimum weighting in a group of stocks or in a sector if they feel this group is likely to outperform the market of their benchmark index. Many times the managers may not have specific stock level alphas or stock specific views, but will apply a higher weighting to the group as a whole. Additionally, managers may specify a maximum weighting for a group of stocks if they believe a particular group will underperform the market and they do not have a view on any particular stock.

*Risk Management.* Portfolio managers may at times be suspicious of the estimated portfolio risk, stock volatility, or covariance across names from a particular risk model. In this case, managers are mostly concerned about type II error—that is, the potential for the risk model to present false positive relationships. A desired property of optimizers is that the results will exploit beneficial relationships. But if these relationships are false positive relationships, the solution will actually increase, rather than decrease, portfolio risk. Thus, as means to provide an added level of safety surrounding potential false positive relationships, managers may specify maximum position sizes, maximum levels of risk exposure, or a maximum stock specific weight (e.g., hold no more than 5 or 10% of the total portfolio value in a specific stock). This constraint is intended to protect the fund from potential errors in the input data.

*Manage Transaction Costs.* The effect of transaction costs and their drag on performance can often be detrimental to fund performance. The larger the position size, the higher the transaction cost. Many times we observe the liquidation cost (selling the order) is much more expensive than the acquisition cost (buying the order). Managers are more likely to buy stocks in favorable market conditions and sell stocks when they have fallen out of favor, when volatility has spiked, and when liquidity dries up. Thus, the liquidation cost is often much more costly than the purchasing